

LSTM Architecture: The Three Gates

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1. Overview

Key point: An LSTM cell uses three gates to manage its two memories. The forget gate removes from the cell state, the input gate adds to it, and the output gate reads from it to make the new hidden state.

The [LSTM intuition Note](#) gave the idea: an LSTM keeps a long-term memory (the **cell state** c_t) next to its short-term memory (the **hidden state** h_t), and a more complex cell lets the two talk to each other. This Note opens that cell. Every part of it is either a small neural network layer or a simple element-by-element operation on vectors.

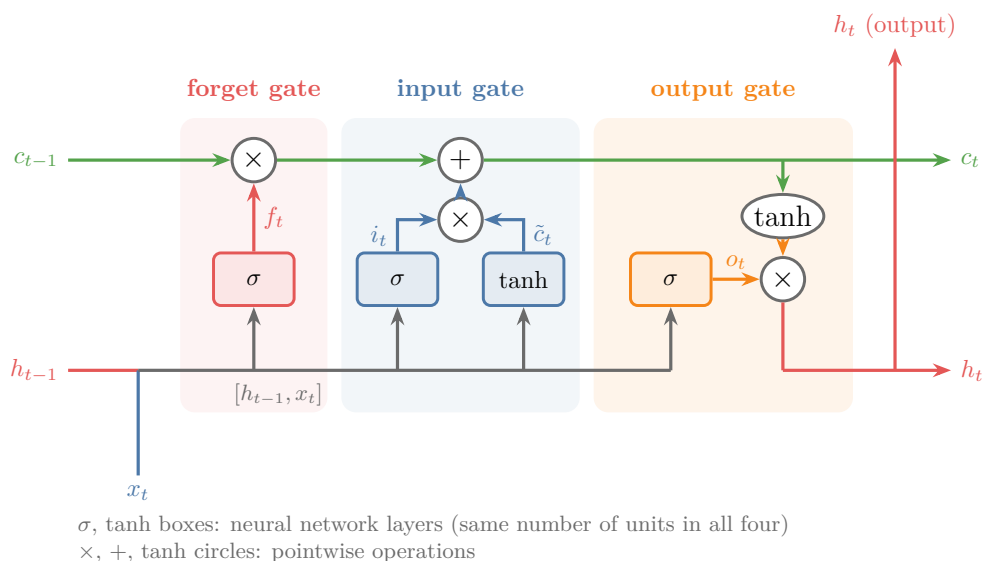


Figure 1: The LSTM cell. The green line is the cell state, the red line the hidden state, blue the current input. The forget gate (red area) scales the old cell state, the input gate (blue area) adds new candidate values, and the output gate (orange area) turns the new cell state into the new hidden state. Boxes are neural network layers; circles are pointwise operations

Figure 1 is the whole cell. By the end of this Note, every symbol in it has a meaning and a formula.

2. Prerequisites

- The [LSTM intuition Note](#): cell state and hidden state, three inputs and two outputs.
- The [RNN forward propagation Note](#): time steps, row vectors, the weight matrices of a recurrent layer.
- The [forward propagation Note](#): a layer of nodes computes weights times inputs, plus biases, through an activation.
- The [activation functions Note](#): the sigmoid (output between 0 and 1) and tanh (output between -1 and 1).

3. What the cell must do

Key point: Three inputs (c_{t-1} , h_{t-1} , x_t), two outputs (c_t , h_t), and three jobs: remove from the cell state, add to the cell state, compute the hidden state. One gate does each job.

As the [LSTM intuition Note](#) showed, the cell at time step t receives the previous cell state c_{t-1} , the previous hidden state h_{t-1} and the current input x_t . It returns the new cell state c_t and the new hidden state h_t . Inside, it does three things, one per gate:

Gate	Job	Area in Figure 1
Forget gate	remove unneeded information from the cell state	red
Input gate	add new important information to the cell state	blue
Output gate	compute the hidden state h_t from the cell state	orange

The forget gate and the input gate together update the cell state, from c_{t-1} to c_t . Both decide on the basis of the current input x_t and the previous hidden state h_{t-1} .

4. The building blocks

Key point: Everything in the cell is a vector of the same length (the number of units), except the input x_t . The boxes are neural network layers and the circles are pointwise operations.

4.1 Cell state and hidden state are vectors of the same length

Key point: c_t and h_t are vectors, for example $[0.1, 0.95, 0.6]$, and they always have the same number of entries.

Mathematically, c_t and h_t are vectors: lists of numbers. If h_t has 3 numbers, such as $[0.1, 0.95, 0.6]$, then c_t also has exactly 3 numbers (different ones). The rule never breaks in an LSTM. The length is the number of **units** of the layer, a hyperparameter we choose.

4.2 The input is a vector of any length

Key point: x_t is the current word turned into a vector. Its length has nothing to do with the number of units.

Take a sentiment task whose reviews use only three words: cat, mat, rat. With one-hot encoding, cat is $[1, 0, 0]$, mat is $[0, 1, 0]$ and rat is $[0, 0, 1]$. The review “cat mat rat” has 3 time steps, and at each one the cell receives one word as x_t , exactly as in the [RNN forward](#)

propagation Note. Any vectorisation works (one-hot, bag of words, TF-IDF, word2vec), and the length of x_t can be larger, smaller or equal to the number of units.

4.3 Four more vectors inside

Key point: f_t (forget gate), i_t and $\tilde{c}_t$ (input gate) and o_t (output gate) all have the same length as c_t and h_t .

The cell computes four more vectors:

- f_t in the forget gate;
- i_t and $\tilde{c}_t$ in the input gate; $\tilde{c}_t$ is the **candidate cell state**;
- o_t in the output gate.

All six vectors, c_t , h_t , f_t , i_t , $\tilde{c}_t$ and o_t , have the same length. Section 4.5 shows why.

4.4 Pointwise operations

Key point: The circles in Figure 1 work element by element: multiply, add, or apply $\tanh$ to each entry separately. The result has the same length as the inputs.

A **pointwise operation** (also called element-wise) works on each position of a vector separately. Multiplication and addition take two vectors of the same length; $\tanh$ takes one vector. We write pointwise multiplication as $\odot$.

1. **In words:** combine the first entries, then the second entries, and so on.
2. **Formula:**

$$[a_1, a_2, a_3] \odot [b_1, b_2, b_3] = [a_1 b_1, a_2 b_2, a_3 b_3], \quad \tanh([a_1, a_2, a_3]) = [\tanh a_1, \tanh a_2, \tanh a_3]$$

3. **Example:** with $a = [4, 5, 6]$ and $b = [1, 2, 3]$:

$$a \odot b = [4, 10, 18], \quad a + b = [5, 7, 9], \quad \tanh(a) = [0.9993, 0.9999, 1.0000]$$

$\tanh$ squeezes every entry into the range -1 to 1 , so large values all land close to 1 .

4.5 The boxes are neural network layers

Key point: Each of the four boxes is a fully connected layer: three with a sigmoid activation, one with $\tanh$. All four have the same number of units, and that number is the length of every vector in section 4.3.

Each box in Figure 1 is an ordinary layer of nodes, like a hidden layer of an ANN. Every node computes a weighted sum plus a bias and applies an activation function:

- the forget gate's layer, the input gate's i_t layer and the output gate's layer use the **sigmoid**;
- the candidate layer that makes $\tilde{c}_t$ uses **tanh**.

The number of nodes per layer is a hyperparameter, such as 3 or 128. Whatever we choose, all four layers get the same number. Each layer outputs one number per node, so f_t , i_t , $\tilde{c}_t$ and o_t have as many entries as there are units, and so do c_t and h_t .

5. The forget gate

Key point: A sigmoid layer looks at h_{t-1} and x_t and outputs f_t , a number between 0 and 1 for every entry of the cell state. Multiplying c_{t-1} by f_t keeps that fraction of each entry: 1 keeps everything, 0 erases it.

5.1 Computing f_t

Key point: Join h_{t-1} and x_t into one vector, pass it through a fully connected sigmoid layer: $f_t = \sigma([h_{t-1}, x_t] W_f + b_f)$.

Take 3 units and a 4-number input x_t . Then h_{t-1} and c_{t-1} have 3 numbers each, because their length equals the number of units.

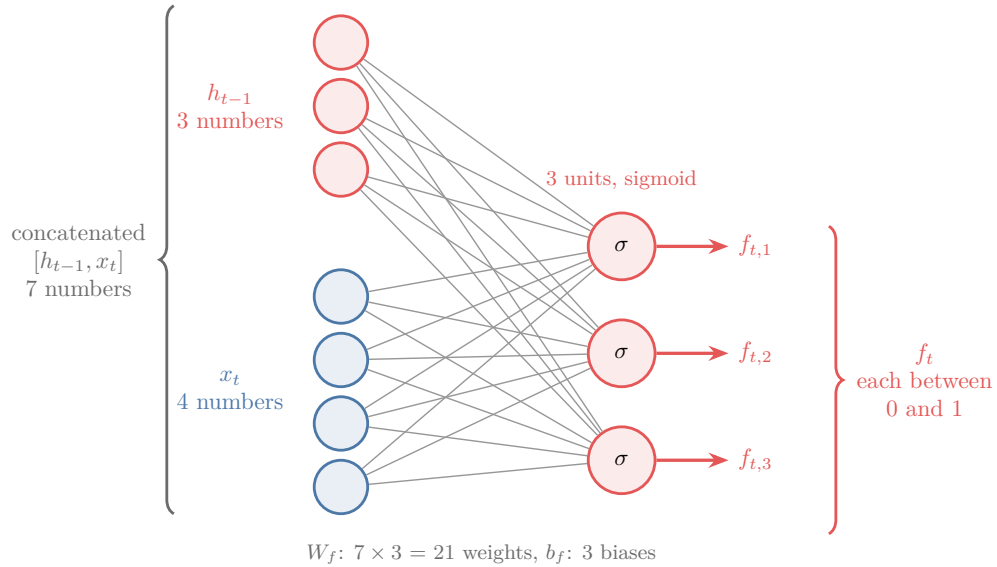


Figure 2: The forget gate's layer. The 3 numbers of h_{t-1} and the 4 numbers of x_t are joined into one input of 7 numbers, fully connected to 3 sigmoid units. Their 3 outputs form f_t

Figure 2 draws the layer. Joining two vectors end to end is **concatenation**, written $[h_{t-1}, x_t]$: here $3 + 4 = 7$ numbers. The 7 inputs connect to all 3 nodes, so the layer has $7 \times 3 = 21$ weights, collected in the matrix W_f , and 3 biases b_f .

1. **In words:** concatenate the previous hidden state and the current input, multiply by the forget gate's weights, add its biases and apply the sigmoid.

2. **Formula:**

$$f_t = \sigma([h_{t-1}, x_t] W_f + b_f)$$

Shapes: $(1 \times 7)(7 \times 3) = 1 \times 3$, plus the 1×3 bias, and the sigmoid keeps 1×3 . So f_t has 3 numbers, the same as c_{t-1} .

3. **Example:** section 8 works through all the gates with numbers.

5.2 Removing from the cell state

Key point: $f_t \odot c_{t-1}$ is the old cell state with each entry scaled by its gate value.

The second step of the forget gate multiplies the old cell state, pointwise, by f_t . The result $f_t \odot c_{t-1}$ moves on along the cell-state line.

1. **In words:** keep, from each entry of the old cell state, the fraction given by the matching entry of f_t .

2. **Formula:**

$$f_t \odot c_{t-1}$$

3. **Example:** with $c_{t-1} = [4, 5, 6]$:

- $f_t = [0.5, 0.5, 0.5]$ gives $[2, 2.5, 3]$: half of the memory is forgotten;
- $f_t = [1, 1, 1]$ gives $[4, 5, 6]$: nothing is forgotten;
- $f_t = [0, 0, 0]$ gives $[0, 0, 0]$: everything is erased.

A gate lets something through or stops it. Because the sigmoid keeps every entry of f_t between 0 and 1, f_t decides how much of each entry of c_{t-1} passes, anywhere from 0% to 100%. And f_t itself is decided by the current input and the previous hidden state. In the story of the [LSTM intuition Note](#), the forget gate is what removes a king from memory once the story reveals his death.

Extra: The first LSTM (Hochreiter and Schmidhuber 1997) had no forget gate: it had only input and output gates, and its cell state could only accumulate. Gers, Schmidhuber and Cummins (2000) added the forget gate, so that a cell can learn to reset itself at the right moments; without resets, the state could grow without limit on long continuous input streams and break the network down. Goodfellow §10.10.1 calls this context-dependent self-loop weight “a crucial addition”. The LSTM used today, in Keras and elsewhere, includes the forget gate.

6. The input gate

Key point: The input gate proposes new values ($\tilde{c}_t$, from a tanh layer), filters them (i_t , from a sigmoid layer), and adds what passes to the cell state: $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$.

The input gate adds new, important information to the cell state, in three stages. In the story, it is what writes the new king into memory.

6.1 Stage 1: the candidate cell state

Key point: A tanh layer turns $[h_{t-1}, x_t]$ into candidate values, each between -1 and 1 .

The **candidate cell state** $\tilde{c}_t$ holds the new values the cell might add, based on the current input and the previous hidden state. It comes from a tanh layer with its own weights W_c (7×3 in our example) and biases b_c :

$$\tilde{c}_t = \tanh\left([h_{t-1}, x_t] W_c + b_c\right)$$

6.2 Stage 2: the filter i_t

Key point: A sigmoid layer decides, entry by entry, how much of each candidate value gets in.

Not every candidate value deserves a place in the long-term memory. A sigmoid layer with weights W_i and biases b_i computes the filter i_t :

$$i_t = \sigma\left([h_{t-1}, x_t] W_i + b_i\right)$$

The pointwise product $i_t \odot \tilde{c}_t$ is the filtered candidate. With $\tilde{c}_t = [4, 5, 6]$ for illustration and $i_t = [0.5, 0.5, 0.5]$, it is $[2, 2.5, 3]$: half of each candidate value goes in. An i_t of 1 lets all of it in, 0 lets none in.

6.3 Stage 3: the new cell state

Key point: Add what the forget gate kept and what the input gate lets in.

1. **In words:** the new long-term memory is the old memory after forgetting, plus the filtered new information.
2. **Formula:**

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

3. **Example:** with $f_t \odot c_{t-1} = [2, 2.5, 3]$ and $i_t \odot \tilde{c}_t = [0.1, 0, 0.3]$:

$$c_t = [2.1, 2.5, 3.3]$$

c_t leaves the cell along the green line and becomes c_{t-1} for the next time step.

6.4 How the cell state carries information far

Key point: If $f_t = 1$ and $i_t = 0$, then $c_t = c_{t-1}$ exactly. The cell state can carry a value unchanged across any number of time steps.

The trouble with a simple RNN is that information from early words fades as it is passed along the chain (see the [problems with RNNs Note](#)). The cell-state line avoids that. Suppose

$c_{t-1} = [4, 5, 6]$, the forget gate is fully open, $f_t = [1, 1, 1]$, and the input gate is closed, $i_t = [0, 0, 0]$:

$$c_t = [1, 1, 1] \odot [4, 5, 6] + [0, 0, 0] \odot \tilde{c}_t = [4, 5, 6]$$

Nothing is lost. If the cell decides at every step that nothing should be removed and nothing added, the information from the beginning of a long sentence reaches its end intact. The gates decide, step by step, how much of the cell state moves on.

Extra: Goodfellow §10.10.1 describes the cell state as having a linear self-loop whose weight is the forget gate: introducing such self-loops “to produce paths where the gradient can flow for long durations” is the core contribution of the LSTM (Hochreiter and Schmidhuber 1997). The original paper reports bridging time lags of more than 1000 discrete time steps on artificial tasks. The [LSTM intuition Note](#) tests the effect on real reviews.

7. The output gate

Key point: Squash the new cell state with $\tanh$, then filter it with o_t : $h_t = o_t \odot \tanh(c_t)$.

The output gate computes the hidden state h_t . h_t goes to the next time step as the short-term memory, and it can also be the cell’s output at this step, depending on the task. So the hidden state is read out of the cell state. The output gate works in two steps.

1. **Squash:** apply $\tanh$ pointwise to c_t , bringing every entry between -1 and 1 .
2. **Filter:** a sigmoid layer with weights W_o and biases b_o computes o_t from $[h_{t-1}, x_t]$, and multiplies it pointwise with $\tanh(c_t)$.

$$o_t = \sigma([h_{t-1}, x_t] W_o + b_o), \quad h_t = o_t \odot \tanh(c_t)$$

Shapes: o_t and $\tanh(c_t)$ are both 1×3 , so h_t is 1×3 , the same as h_{t-1} , ready for the next time step.

8. One time step with numbers

Key point: Six formulas make one LSTM time step. With small hand-picked weights we can follow every number, and Keras gives the same result.

1. **In words:** from h_{t-1} and x_t , compute the three gates and the candidate; update the cell state; read the hidden state out of it.
2. **Formula:**

$$\begin{aligned} f_t &= \sigma([h_{t-1}, x_t] W_f + b_f) & i_t &= \sigma([h_{t-1}, x_t] W_i + b_i) \\ \tilde{c}_t &= \tanh([h_{t-1}, x_t] W_c + b_c) & o_t &= \sigma([h_{t-1}, x_t] W_o + b_o) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t & h_t &= o_t \odot \tanh(c_t) \end{aligned}$$

3. **Example:** 2 units, vocabulary cat, mat, rat, all biases 0. The cell has $h_{t-1} = [0.3, -0.2]$ and $c_{t-1} = [0.8, -0.5]$, and reads “mat”, $x_t = [0, 1, 0]$. So $[h_{t-1}, x_t] = [0.3, -0.2, 0, 1, 0]$. Each weight matrix has 5 rows (for $h_1, h_2, \text{cat}, \text{mat}, \text{rat}$) and 2 columns:

$$W_f = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \\ 1 & 0 \\ 2 & -1 \\ 0 & 1 \end{bmatrix} \quad W_i = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \\ 0.5 & 0.5 \\ -1 & 2 \\ 0 & 0 \end{bmatrix} \quad W_c = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \\ 0.3 & 0.3 \\ 0.5 & -1 \\ 0 & 0 \end{bmatrix} \quad W_o = \begin{bmatrix} 0.3 & 0 \\ 0 & 0.3 \\ 0 & 0 \\ 1 & 0.5 \\ 0 & 0 \end{bmatrix}$$

Multiplying $[0.3, -0.2, 0, 1, 0]$ by a matrix gives 0.3 times its first row, minus 0.2 times its second row, plus its “mat” row.

- **Forget gate:** $[0.15, 0] + [0, -0.1] + [2, -1] = [2.15, -1.10]$, so $f_t = \sigma([2.15, -1.10]) = [0.896, 0.250]$.
- **Input gate:** $[0.06, 0] + [0, -0.04] + [-1, 2] = [-0.94, 1.96]$, so $i_t = [0.281, 0.877]$.
- **Candidate:** $[0.03, 0] + [0, -0.02] + [0.5, -1] = [0.53, -1.02]$, so $\tilde{c}_t = \tanh(\cdot) = [0.485, -0.770]$.
- **Output gate:** $[0.09, 0] + [0, -0.06] + [1, 0.5] = [1.09, 0.44]$, so $o_t = [0.748, 0.608]$.
- **Cell state:**

$$f_t \odot c_{t-1} = [0.896 \times 0.8, 0.250 \times (-0.5)] = [0.717, -0.125]$$

$$i_t \odot \tilde{c}_t = [0.281 \times 0.485, 0.877 \times (-0.770)] = [0.136, -0.675]$$

$$c_t = [0.717 + 0.136, -0.125 - 0.675] = [0.853, -0.800]$$

- **Hidden state:**

$$\tanh(c_t) = [0.693, -0.664], \quad h_t = [0.748, 0.608] \odot [0.693, -0.664] = [0.518, -0.404]$$

Unit 1 kept most of its old memory ($f = 0.896$) and added a little. Unit 2 forgot most of its old value ($f = 0.250$) and let in most of a strongly negative candidate ($i = 0.877$). Keras’ LSTM layer, given these weights and starting states, returns the same c_t and h_t (Notebook).

9. Counting the parameters

Key point: Four layers, each with $(\text{units} + \text{input features}) \times \text{units weights}$ and units biases: $4((u + d)u + u)$. For 3 units and 4 input features, 96 parameters, four times a SimpleRNN’s 24.

With u units and input vectors of d numbers, each of the four layers (W_f, W_i, W_c, W_o) has $(u + d) \times u$ weights and u biases.

1. **In words:** count one layer’s weights and biases, then multiply by four.
2. **Formula:**

$$\text{parameters} = 4((u + d)u + u)$$

3. **Example:** $u = 3$ units and $d = 4$ input features, as in Figure 2:

$$4 \left((3 + 4) \times 3 + 3 \right) = 4 \times (21 + 3) = 96$$

Keras counts 96 for LSTM(3) on 4 input features, and 24 for SimpleRNN(3) on the same input: a simple RNN has one such layer, an LSTM four.

Python: An LSTM layer in Keras, and where its weights live.

```
model = keras.Sequential([
    keras.Input(shape=(None, 4)),    # any number of time steps
    keras.layers.LSTM(3)])
model.count_params()                # 96
kernel, recurrent_kernel, bias = model.layers[0].get_weights()
# kernel (4, 12): the x_t rows of all four layers
# recurrent_kernel (3, 12): the h_(t-1) rows
# bias (12,); columns in the order i, f, c~, o
```

Extra: Keras does not store one 7×3 matrix per gate. It splits every gate's matrix into its x_t rows (the **kernel**) and its h_{t-1} rows (the **recurrent_kernel**), and places the four gates side by side in the order input, forget, candidate, output. Multiplying the concatenation $[h_{t-1}, x_t]$ by the stacked matrix is the same as h_{t-1} times the top rows plus x_t times the bottom rows. Keras also starts the forget gate's biases at 1 instead of 0 (**unit_forget_bias=True**), following Jozefowicz et al. (2015), as stated in the Keras documentation: the forget gate then starts mostly open, $\sigma(1) = 0.73$.

10. The gates on a real review

Key point: In a trained LSTM, every gate value is computed fresh for every word. On a real review, the forget gate stays mostly open, while the cell state shifts at the words that carry sentiment.

To see the gates at work, we train a small LSTM on movie reviews from the IMDB dataset (keras.datasets): an embedding layer (16 numbers per word), an LSTM(8) layer and a sigmoid output, on 10,000 reviews cut to their last 200 words: 8,000 for training and 2,000 held back for validation. After 3 epochs it classifies 86% of the validation reviews correctly. Then we run one real test review through the trained cell by hand, word by word, and record f_t , i_t , o_t and c_t ; the hand computation gives exactly Keras' prediction (Notebook). We chose, among the 21 test reviews of 10 to 30 known words, the one whose running prediction swings the most.

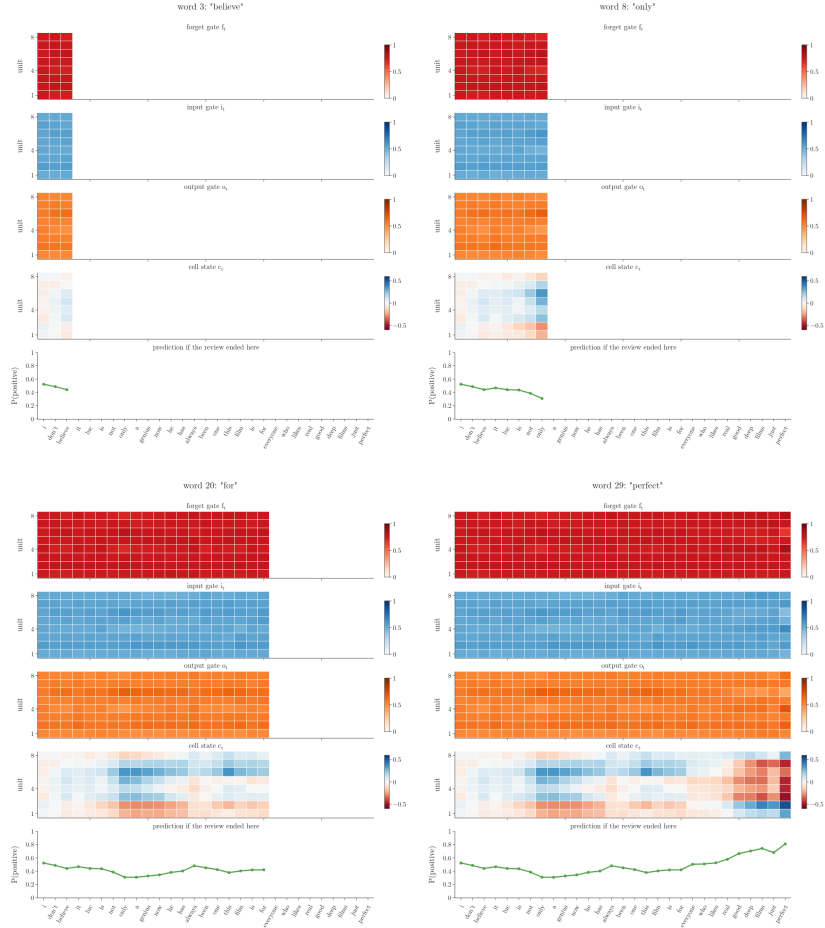


Figure 3: The trained LSTM reading a real IMDB review, one word per frame. Top four panels: the forget, input and output gates (0 to 1) and the cell state of the 8 units. Bottom: the prediction the model would give if the review ended at that word

Figure 3 shows the review “i don’t believe it luc is not only a genius now he has always been one this film is for everyone who likes real good deep films just perfect”, labelled positive.

- **The forget gate stays mostly open.** Its values lie between 0.65 and 0.86 (mean 0.76) for every unit and every word, so each step keeps most of the cell state and drops a part of it.
- **The cell state moves with the meaning.** Around “not only”, several units of c_t turn clearly positive or negative, and the prediction falls from 0.52 to 0.31. From “good deep films” to “perfect” other units swing strongly, and the prediction climbs to 0.81.
- **The prediction comes from the cell state.** The output layer sees only $h_t = o_t \odot \tanh(c_t)$, so the shifts in c_t are what move the prediction.

The gates of this small model vary only a little from word to word; the large changes come from the cell state, which adds up the filtered candidates over many words.

11. Summary

Part	Formula	Activation	Job
Forget gate	$f_t = \sigma([h_{t-1}, x_t]W_f + b_f)$	sigmoid	how much of each entry of c_{t-1} to keep
Input gate	$i_t = \sigma([h_{t-1}, x_t]W_i + b_i)$	sigmoid	how much of each candidate value to add
Candidate cell state	$\tilde{c}_t = \tanh([h_{t-1}, x_t]W_c + b_c)$	tanh	the new values that could be added
Output gate	$o_t = \sigma([h_{t-1}, x_t]W_o + b_o)$	sigmoid	how much of $\tanh(c_t)$ to show
Cell state	$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$	none	the long-term memory
Hidden state	$h_t = o_t \odot \tanh(c_t)$	tanh	the short-term memory and output

- All vectors inside the cell have length equal to the number of units; only x_t can differ.
- The four boxes are fully connected layers on the concatenation $[h_{t-1}, x_t]$; the circles are pointwise operations.
- The forget gate removes from the cell state, the input gate adds to it, the output gate reads the hidden state out of it.
- With $f_t = 1$ and $i_t = 0$, the cell state passes through unchanged, so information can travel far.
- An LSTM layer has $4((u + d)u + u)$ parameters, four times a SimpleRNN with the same sizes.

12. Sources

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- Keras API documentation: IMDB movie review sentiment classification dataset, keras.io/api/datasets/imdb.

13. Key terms

Term	Meaning
Cell state (c_t)	The LSTM's long-term memory, a vector passed along the top line of the cell
Hidden state (h_t)	The LSTM's short-term memory and output at time t
Units	The number of nodes in each of the four layers; the length of c_t , h_t and every gate vector
Forget gate (f_t)	Sigmoid layer whose output scales each entry of c_{t-1} , removing information
Input gate (i_t)	Sigmoid layer whose output filters the candidate values before they are added
Candidate cell state ($\tilde{c}_t$)	Output of the tanh layer: new values that could be added to the cell state
Output gate (o_t)	Sigmoid layer whose output filters $\tanh(c_t)$ to give h_t
Gate	A sigmoid output between 0 and 1 that sets what fraction of a value passes
Pointwise operation	An operation done entry by entry on vectors of the same length, such as $\odot$
Concatenation ($[h_{t-1}, x_t]$)	Joining two vectors end to end into one longer vector